

Jayesh Janardan Jadhav - Curriculum Vitae

Experimental Physics | Photonics | Nanofabrication

JAYESH JANARDAN JADHAV

Experimental Physics | Photonics | Thin Films | Nanofabrication

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PROFILE

Experimental physicist with experience in thin-film optical design, vacuum deposition, spectroscopy and scientific modelling. M.Sc. research at IIT Bombay focused on IZO/Ag/IZO transparent conducting multilayers, with an increasing emphasis on photonics and light-matter interaction. Incoming Quantum Nanofabrication Researcher in a National Quantum Mission nanofabrication laboratory; detailed project assignment is being finalised, with a broader direction related to fabrication for quantum-computing hardware.

EDUCATION

Master of Science in Physics | Indian Institute of Technology Bombay | 2024-2026

- Thesis: Simulation and Optical Design of Multilayer Transparent Conducting Oxides
- Research centre: National Centre for Photovoltaic Research and Education (NCPRE)
- Guide: Dr. Dinesh Kabra | Mentor: Mr. Subir Manna
- Selected coursework: Light-Matter Interaction, Photonics and Engineering, Ultrafast Sciences, Quantum Mechanics I and II, Electromagnetic Theory I and II, Statistical Physics, Introduction to Condensed Matter Physics, Electronics and Programming Laboratory.

Bachelor of Science (Honours) in Physics | Lovely Professional University | 2020-2023

- Selected coursework: Applied Optics, Waves and Optics, Quantum Mechanics, Statistical Mechanics, Nuclear and Particle Physics, Mathematical Physics I and II, Analogue and Digital Systems.

RESEARCH EXPERIENCE

Incoming Quantum Nanofabrication Researcher | National Quantum Mission | 2026

- Joining a nanofabrication laboratory supporting quantum-technology research.
- Detailed assignment is being finalised; expected direction is fabrication processes for quantum-computing hardware.

M.Sc. Thesis Research | NCPRE, IIT Bombay | 2024-2026

Simulation and Optical Design of Multilayer Transparent Conducting Oxides

- Investigated IZO/Ag/IZO oxide-metal-oxide transparent electrodes for photovoltaic devices.
- Developed a SETFOS and Python workflow for transmission, reflection, absorption, photon-flux and theoretical current calculations.
- Examined non-bulk optical behaviour in sub-10 nm silver using Drude-based interpretation, plasmonic considerations and realistic film-growth constraints.
- Evaluated optical-electrical trade-offs using sheet resistance and Haacke's figure of merit.
- Treated strong simulated gains as theoretical design limits and analysed why roughness, morphology, interfaces and non-uniformity reduce post-fabrication performance.
- Hands-on exposure: RF magnetron sputtering, metal evaporation, cleanroom processing, vacuum handling, UV-Vis-NIR spectroscopy, four-point-probe measurements and profilometry.

Summer Practical Training | Bhabha Atomic Research Centre | May-Jul 2025

High-Pressure Raman Spectroscopy

- Investigated $\text{Na}_2\text{Cu}_2\text{TeO}_6$ under pressures up to 10 GPa using diamond anvil cells and Raman spectroscopy.
- Developed experience with alignment, pressure-dependent measurements and interpretation under extreme conditions.

Experimental Campaign | INDUS-2 Synchrotron, RRCAT | Jun-Jul 2025

- Participated as a BARC team member in Beamline 11 extreme-condition X-ray diffraction measurements.
- Collected data across multiple samples and pressure conditions in a coordinated facility campaign.

SELECTED PROJECTS

PANI/ Fe_3O_4 Electromagnetic-Interference Shielding Composite | 2022-2023

- Synthesised magnetite and an in-situ polyaniline composite.
- Connected FESEM, XRD and FTIR characterization with vector-network-analyser shielding measurements.

Microbial Fuel Cell | 2022

- Supported assembly and chemical preparation for a bio-waste-based microbial fuel-cell project exploring hydrogen production.

Random Signal Generator without Microprocessor | IIT Bombay | 2024

- Designed a light-and-sound random signal generator as an electronics course project for rural animal deterrence.

Scientific Vector Diagrams | 2023

- Created scientific diagrams for colleagues' thesis projects using Adobe Illustrator and Affinity Designer.

IIT Bombay M.Sc. Physics Fresher Guide | 2026

- Designed a digital guide for incoming students, combining information architecture, web design and student coordination.

TECHNICAL SKILLS

Fabrication and laboratory: RF magnetron sputtering, thermal/metal evaporation, cleanroom workflow, vacuum handling, diamond anvil cells.

Characterization: UV-Vis-NIR spectroscopy, Raman spectroscopy, four-point probe, profilometry, synchrotron XRD exposure.

Modelling and computation: SETFOS, multilayer optical analysis, Drude-model interpretation, AM1.5G photon-flux calculations, Python.

Design and communication: Adobe Illustrator, Affinity Designer, scientific diagrams, technical writing and presentations.

LEADERSHIP

M.Sc. Representative | IIT Bombay | 2025-2026

- Represented and coordinated two M.Sc. Physics batches totalling approximately 120 students.

Class Representative | IIT Bombay | 2024-2025

- Represented a class of approximately 60 first-year M.Sc. Physics students and coordinated

student activities.

Incredible 8 Team Lead | Lovely Professional University | 2022

- Led an eight-person departmental team in a multi-phase community support project.

AWARDS AND ACADEMIC DEVELOPMENT

- IIT JAM 2024: All India Rank 84.

- CUET-PG 2024: Score 159.

- Inter IIT 7.0 Photography: 5th position as a member of the IIT Bombay photography team, 2024.

- PG-Cult IIT Bombay Photography Competition: 3rd prize, 2024.

- Summer School on Optics and Photonics, IIT Delhi, 2025.

- Fourth International Conference on Recent Advances in Fundamental and Applied Sciences (RAFAS), 2023.

- Workshop on Testing Aspects of General Relativity, IIT Gandhinagar, 2023.

INTERESTS

Nanophotonics, light-matter interaction, optical instrumentation, quantum-device fabrication, experimental condensed matter physics, photography, fine art, swimming and trekking.

LANGUAGES

Marathi, Hindi and English.